

RAY VERMA

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EDUCATION

New York University

B.S., double major in Computer Science and Mathematics

Aug. 2023 – Jul. 2027

Abu Dhabi, UAE

GPA: 3.95/4.00. Selected coursework: Algorithms, Data Structures, Linear Algebra, Multivariable Calculus, Deep Learning, Predictive Analytics. Global study in Abu Dhabi, New York, Paris and Los Angeles.

RESEARCH EXPERIENCE

Center for AI and Robotics, NYU | *Research Assistant*

Feb. 2026 – Present

Abu Dhabi, UAE

- Decentralized morphogenesis: Developing local controllers for homogeneous robot swarms that form, reconfigure and recover global shapes using only neighborhood observations.
- Dynamic target assignment: Designing GoalNCA, a neural cellular automata approach for decentralized frontier target acquisition and scalable lattice formation control.
- Learned Morphogenesis: Developing NavCA, an end-to-end equivariant graph cellular automata controller that generalizes across target shapes without global coordinates.

AGI Lab, Stanford University | *Research Assistant*

Aug. 2025 – Present

Remote / Stanford, CA

- Geometry-aware representations: Developed Random Voronoi Networks, interpretable feature maps that induce geometric bias in simple models and improve F1 performance by up to 40%.
- Unsupervised dataset embeddings: Developed Expand and Reduce, a schema-agnostic collection-level representation for AutoML that outperformed supervised benchmarks by 18%.
- Tabular foundation models: Investigating Prior-Fitted Networks for data-efficient prediction and transfer from synthetic priors to heterogeneous real-world datasets.

Genetic Heritage Group, NYU | *Research Assistant*

Aug. 2024 – Sep. 2026

Abu Dhabi, UAE

- Population-scale microbiome analysis: Built an archetypal-analysis and visualization toolkit with a 20× faster dimensionality-reduction workflow, processing more than 30,000 samples.
- Causal interaction discovery: Adapted HaLLA and PCMCi+ for longitudinal bacteria-virus analysis, identifying ecological dependencies relevant to drug discovery for antimicrobial-resistant pathogens.
- Cross-dataset validation: Collaborated with UC Santa Cruz across four microbiome datasets to evaluate community structure and cumulative biological effects under a common workflow.

INDUSTRY EXPERIENCE

Galatea Bio | *AI Engineer Intern*

Jun. 2025 – Aug. 2025

Remote / Miami, FL

- Memory-aware scheduling: Engineered RAM-efficient execution for chromosome-level genomic workflows using symbolic regression to predict task demand and applied conformal bounds to limit over-scheduling.
- Adaptive task packing: Published and presented (AAAI) a resource-aware orchestration methods that reduced multi-task runtime by 60% while increasing system utilization by 75%.
- Workflow reliability: Validated scheduling policies through simulation and visualization, refactored core components and hardened execution with Prefect and GitHub Actions.

Emerton Data | *AI Engineer Intern*

Jan. 2025 – May 2025

Paris, France

- Low-resource reasoning: Fine-tuned Qwen and Llama models for chain-of-thought translation with fewer than 10,000 samples using GRPO reinforcement learning.
- Multi-reward optimization: Designed a dynamically weighted objective over seven reward functions and a reproducible 10-day remote-GPU training pipeline for a Government of Monaco project.
- Reproducible experimentation: Instrumented training with Hugging Face, Unsloth and Weights & Biases to compare reward schedules, checkpoints and stability.